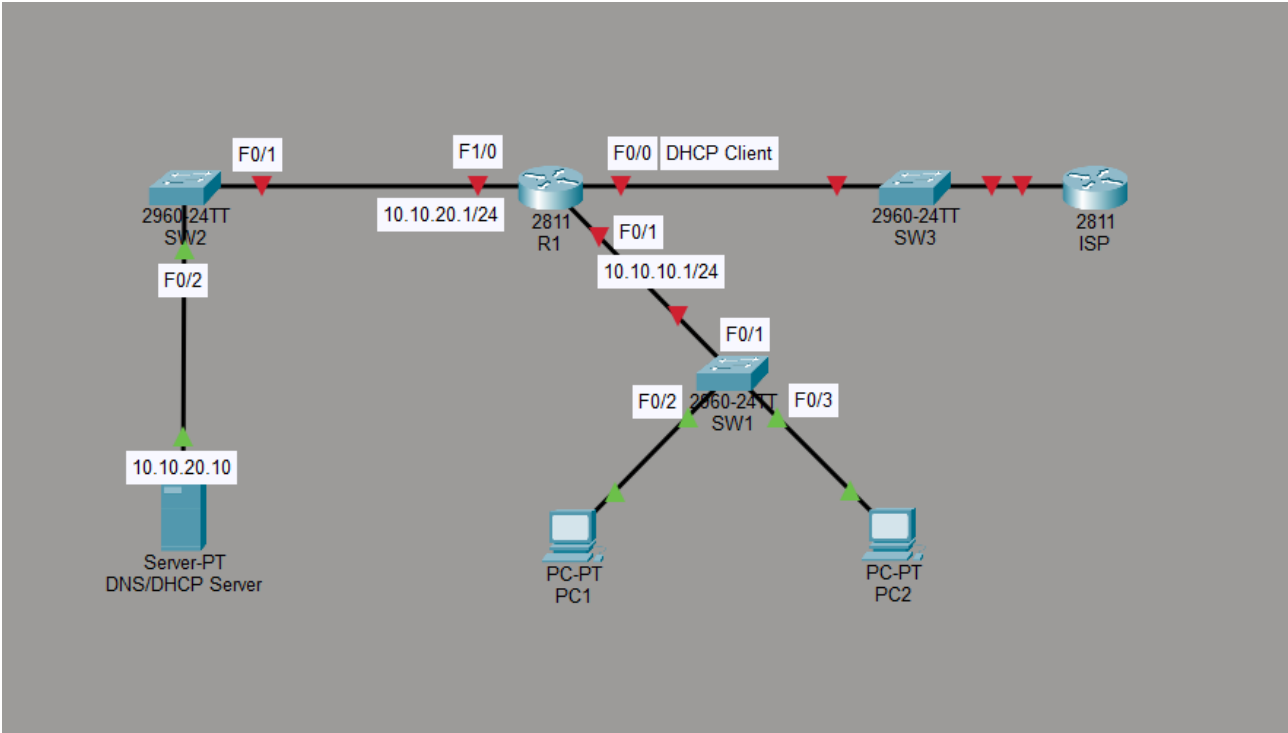


NET512 — Weekly Practical

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STUDENT NO.	20261193	DATE	02 October 2026

Topology



Task 97 — DHCP Client

The first screenshot shows the configuration of an ISP router. The CLI window displays the following commands and output:

```
ISP#
ISP(config-if)#ip address 209.165.201.1 255.255.255.0
ISP(config-if)#no shutdown

ISP(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
ISP(config)#ip ip dhcp excluded-address 209.165.201.1 209.165.201.9
% Invalid input detected at '^' marker.

ISP(config)#ip dhcp excluded-address 209.165.201.1 209.165.201.9
ISP(config)#ip dhcp pool ISP-POOL
ISP(dhcp-config)#network 209.165.201.0 255.255.255.0
ISP(dhcp-config)#default-router 209.165.201.1
ISP(dhcp-config)#end
ISP#
%SYS-5-CONFIG_I: Configured from console by console

ISP#show running-config | section dhcp
ip dhcp excluded-address 209.165.201.1 209.165.201.9
ip dhcp pool ISP-POOL
 network 209.165.201.0 255.255.255.0
 default-router 209.165.201.1
ISP#
```

The second screenshot shows the configuration of a DHCP client on R1. The CLI window displays the following commands and output:

```
R1#
R1(config)#int fa0/0
R1(config-if)#description External - DHCP Client
R1(config-if)#ip address dhcp
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R1(config-if)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show ip interface brief
%DHCP-6-ADDRESS_ASSIGN: Interface FastEthernet0/0 assigned DHCP address 209.165.201.10, mask 255.2
R1#
R1#show ip interface brief


| Interface       | IP-Address     | OK? | Method | Status                | Protocol |
|-----------------|----------------|-----|--------|-----------------------|----------|
| FastEthernet0/0 | 209.165.201.10 | YES | DHCP   | up                    | up       |
| FastEthernet0/1 | 10.10.10.1     | YES | manual | up                    | up       |
| FastEthernet1/0 | 10.10.20.1     | YES | manual | up                    | up       |
| FastEthernet1/1 | unassigned     | YES | unset  | administratively down | down     |
| Vlan1           | unassigned     | YES | unset  | administratively down | down     |


R1#
```

The third screenshot shows the DHCP lease details on R1. The CLI window displays the following commands and output:

```
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show ip interface brief
%DHCP-6-ADDRESS_ASSIGN: Interface FastEthernet0/0 assigned DHCP address 209.165.201.10, mask 255.2
R1#
R1#show ip interface brief


| Interface       | IP-Address     | OK? | Method | Status                | Protocol |
|-----------------|----------------|-----|--------|-----------------------|----------|
| FastEthernet0/0 | 209.165.201.10 | YES | DHCP   | up                    | up       |
| FastEthernet0/1 | 10.10.10.1     | YES | manual | up                    | up       |
| FastEthernet1/0 | 10.10.20.1     | YES | manual | up                    | up       |
| FastEthernet1/1 | unassigned     | YES | unset  | administratively down | down     |
| Vlan1           | unassigned     | YES | unset  | administratively down | down     |


R1#show dhcp lease
Temp IP addr: 209.165.201.10 for peer on Interface: FastEthernet0/0
Temp sub net mask: 255.255.255.0
  DHCP Lease server: 209.165.201.1 , state: Bound
  DHCP Transaction id: B4784DA4
  Lease: 86400 secs, Renewal: 43200 secs, Rebind: 75600 secs
Temp default-gateway addr: 209.165.201.1
Next timer fires after: 11:58:01
Retry count: 0 Client-ID:cisco-00D0.FFC8.D901-Fa0/0
Client-ID hex dump: 63697363662D303044302E46464643382E
                    43930312D4661302F30
  Hostname: R1
R1#
```

Task 98 — DHCP Server

The top screenshot shows the configuration of a DHCP server on a router (R1) via the CLI. The configuration includes creating a pool named LAN-POOL, setting the network to 10.10.10.0/24, excluding the router's interface address (10.10.10.1), and setting the default gateway to 10.10.10.1. The DNS server is set to 10.10.20.10, and the domain name is w10.local. A lease time of 7 days is configured. The configuration is verified using the 'show running-config' command, which displays the DHCP pool configuration. The 'show ip dhcp binding' command shows a successful binding for the IP address 10.10.10.21 to the hardware address 0002.17C6.64E2.

```
R1(config)#ip dhcp pool LAN-POOL
R1(dhcp-config)#network 10.10.10.0 255.255.255.0
R1(dhcp-config)#default-router 10.10.10.1
R1(dhcp-config)#dns-server 10.10.20.10
R1(dhcp-config)#domain-name w10.local
R1(dhcp-config)#lease 7
^
% Invalid input detected at '^' marker.

R1(dhcp-config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show running-config | section dhcp
ip dhcp excluded-address 10.10.10.1 10.10.10.20
ip dhcp pool LAN-POOL
 network 10.10.10.0 255.255.255.0
 default-router 10.10.10.1
 dns-server 10.10.20.10
 domain-name w10.local
ip address dhcp
R1#show ip dhcp binding
IP address      Client-ID/      Lease expiration        Type
               Hardware address
10.10.10.21     0002.17C6.64E2   --                       Automatic
R1#
```

The bottom screenshot shows the IP Configuration window for PC1. The interface FastEthernet0 is configured with DHCP. The IPv4 Address is 10.10.10.21, Subnet Mask is 255.255.255.0, Default Gateway is 10.10.10.1, and DNS Server is 10.10.20.10. The DHCP request was successful. The IPv6 Configuration section shows the Automatic option selected, and the Link Local Address is FE80::202:17FF:FEC6:64E2. The 802.1X section shows the Use 802.1X Security checkbox unchecked.

Interface: FastEthernet0

IP Configuration

- ☒ DHCP ☐ Static DHCP request successful.
- IPv4 Address: 10.10.10.21
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.10.10.1
- DNS Server: 10.10.20.10

IPv6 Configuration

- ☐ Automatic ☒ Static
- IPv6 Address: /
- Link Local Address: FE80::202:17FF:FEC6:64E2
- Default Gateway:
- DNS Server:

802.1X

- ☐ Use 802.1X Security

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Task 99 — External DHCP Server



DNS/DHCP Server

Physical Config **SERVICES** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP**
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

DHCP

Interface: **FastEthernet0** Service: ☒ On ☐ Off

Pool Name: **LAN-POOL**

Default Gateway: **10.10.10.1**

DNS Server: **10.10.20.10**

Start IP Address: **10** **10** **10** **21**

Subnet Mask: **255** **255** **255** **0**

Maximum Number of Users: **50**

TFTP Server: **0.0.0.0**

WLC Address: **0.0.0.0**

Add **Save** **Remove**

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
LAN-POOL	10.10.10.1	10.10.20.10	10.10.10.21	255.255.255.0	50	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	10.10.20.0	255.255.255.0	512	0.0.0.0	0.0.0.0

R1

```

cisco 2811 (MPC860) processor (revision 0x200) with 60416K/5120K bytes of memory
Processor board ID JAD05190MTZ (4292891495)
4 FastEthernet interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

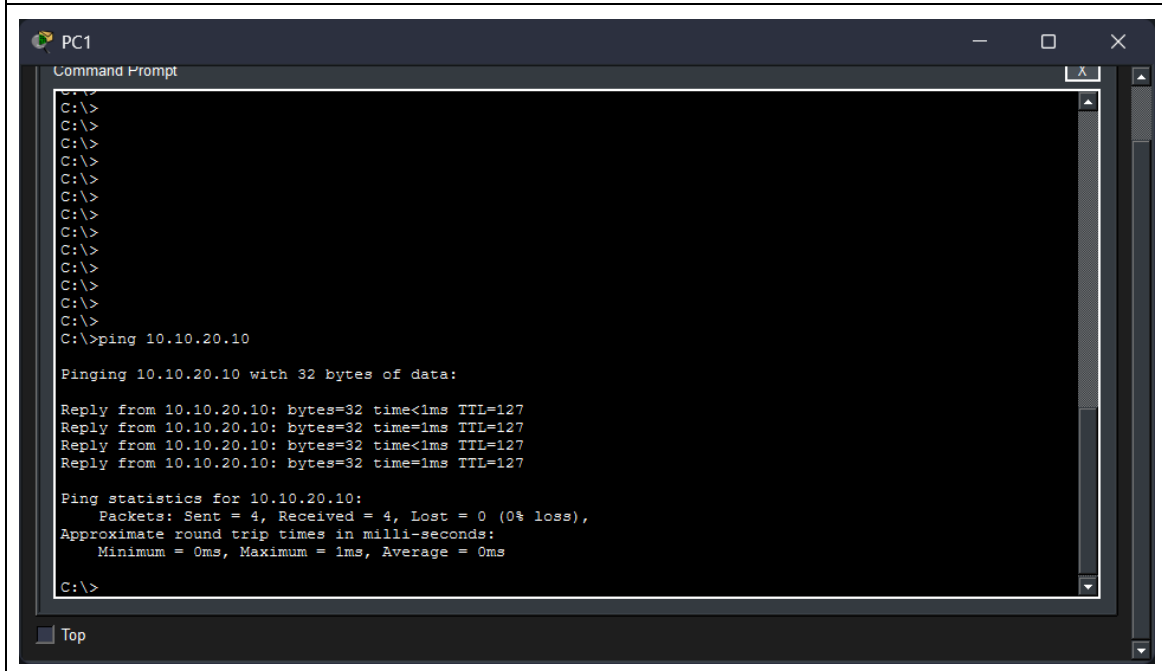
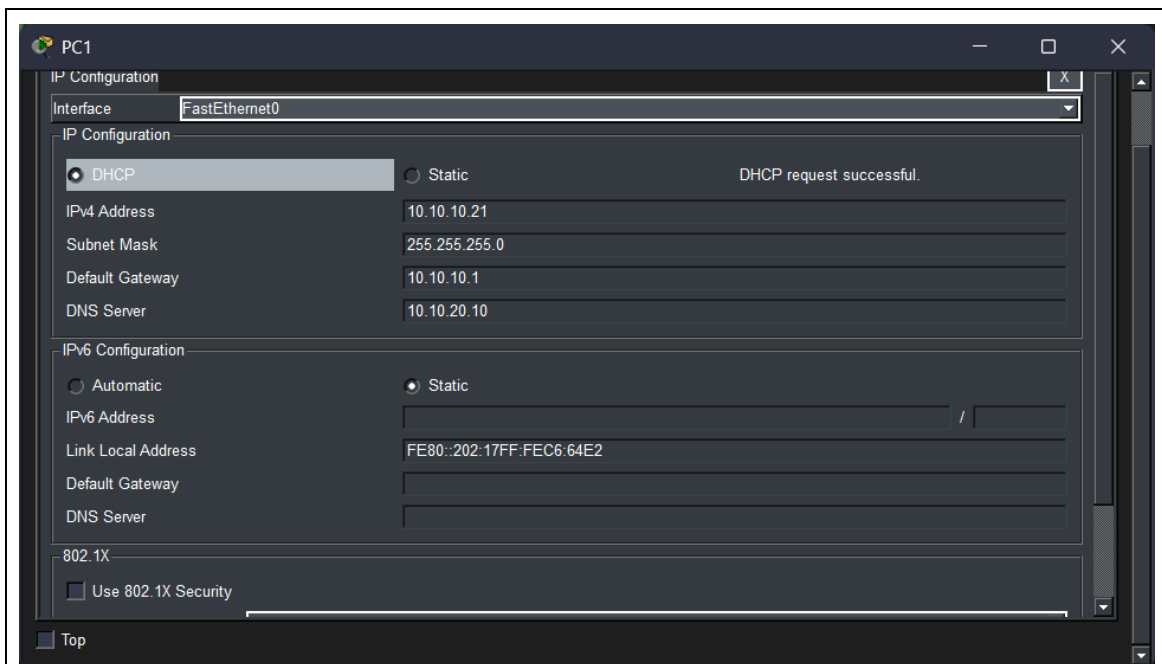
Press RETURN to get started!

R1>enable
R1#conf term
%DHCP-6-ADDRESS_ASSIGN: Interface FastEthernet0/0 assigned DHCP address 209.165.201.10, mask 255.255.255.0,
hostname R1

Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int fa0/1
R1(config-if)#ip helper-address 10.10.20.10
R1(config-if)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
show ip interface fa0/1 | include Helper
    Helper address is 10.10.20.10
R1#
  
```

Copy Paste

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Reflection Questions

Task 100 – IP Helper Address

We need a helper address because of DHCP starting with a broadcast and routers do not forward it. As a result of this, clients that have just booted do not have an address, mask or gateway and thus cannot communicate with specific servers.

The helper address turns the router into a relay on the interface the broadcast arrives from. Intercepting instead of dropping and rewriting it as unicast towards configured server.